

Fate Hardin

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TECHNICAL SKILLS

Languages: JavaScript (ES6+), Python, C++, C#, Java, C, CSS, Swift

Frontend: React, Socket.IO, Electron, HTML/CSS

Backend & Cloud: Node.js, AWS (EC2, Cognito, GameLift, Auto Scaling), Docker, PostgreSQL, SQL

AI & Automation: LangChain, LM Studio, TensorFlow, RAGFlow, N8N, Whisper

DevOps & Infra: EC2 AMIs, Launch Templates, Auto Scaling Groups, WireGuard VPN, Ubuntu Server

WORK EXPERIENCE

XR Technologist | *Brightline Interactive*

January 2024 – January 2025

- Architected a full-stack monitoring platform for a \$4M+ Department of Defense Spatial Computing system — React web client, Electron desktop client, and a bidirectional Socket.IO messaging layer enabling real-time session control and telemetry.
- Engineered CloudXR streaming clients enabling low-latency GPU-rendered XR content delivered from AWS EC2 instances to HTC Vive XR Elite headsets over standard network connections.
- Designed and managed EC2 auto scaling groups, custom AMIs, and launch templates to fully automate VM provisioning — reducing deployment overhead and supporting elastic scaling under variable concurrent XR session load.
- Integrated an LLM with RAGFlow into the Nvidia Omniverse editor as a standalone microservice, enabling context-aware in-editor assistance for 3D environment construction powered by retrieval-augmented generation over project documentation.
- Built and maintained a Magic Leap 2 World Building Plugin for Nvidia Omniverse using Omni Kit, allowing users to spawn, place, and transform 3D primitives in AR space via real-time remote rendering.

Game Engineer | *Immersive Health Group*

February 2020 – December 2023

- Led design and implementation of cloud multiplayer infrastructure using AWS GameLift and Unity Photon, enabling server-authoritative networked experiences across Windows, Mac, and Meta Quest with dynamic session matchmaking.
- Built a collaborative VR medical training environment in Unreal Engine where instructors could observe participants in real time, annotate performance, and review procedure accuracy — used for clinical staff onboarding and compliance training.
- Developed a VR patient intake and IV procedure simulation in UE4, including a fully custom UI system and 3D inventory for equipment selection, used to train staff on proper intravenous administration protocols.
- Owned all UI/UX development across UE4/UE5 and Unity projects, building both diegetic in-world and traditional HUD interfaces for XR and desktop; managed full build and certification pipelines for Meta Quest Store and iOS App Store releases.

PROJECTS

Video-to-Notes Workflow | **Personal**

2024

- Full-stack pipeline: users submit a YouTube playlist URL and receive structured AI-generated notes — React frontend, Socket.IO real-time messaging, Whisper C++ for local speech-to-text, FFMPEG for audio extraction, LLaMA 3.1 via LM Studio for summarization.

Email AI Assistant | **Personal**

2023

- Electron app with a LangChain/OpenAI agent that reads, classifies, and summarizes emails via custom tool definitions. Designed with extensible architecture for future vector DB storage and multi-user support.

Personal Ubuntu Home Server | **Personal**

2023 – Present

- Self-hosted Linux server with WireGuard VPN for secure remote access, Samba cloud drive, Jellyfin media server, and automated iCloud photo backup via icloudpd — fully accessible from any network.

EDUCATION

M.S. in Computer Science — Middle Tennessee State University

Expected May 2028

B.S. in Computer Science — University of Tennessee at Martin

May 2020